

Problem 1. (a) Let $n \geq 3$ be an odd integer that is not a perfect square. Prove that in the ring $\mathbb{Z}[i\sqrt{n}]$ the element 2 is irreducible but is not prime.

(b) Prove that $\mathbb{Z}[i\sqrt{5}]$ is not a unique factorization domain.

Proof. (a) Recall that the norm $N(a + bi\sqrt{n}) = a^2 + b^2n$ is multiplicative: $N(\alpha\beta) = N(\alpha)N(\beta)$, $\forall \alpha, \beta \in \mathbb{Z}[i\sqrt{n}]$.

Suppose 2 is reducible. Then there exist non-zero non-units $\alpha = a + bi\sqrt{n}, \beta = c + di\sqrt{n} \in \mathbb{Z}[i\sqrt{n}]$ such that $2 = \alpha\beta$. Now, If neither α nor β is a unit, then both norms are greater than 1 and the only possibility is $N(\alpha) = N(\beta) = 2$, since $4 = (1)(4) = (2)(2)$. But $n \geq 3 \implies b^2n \geq 3 \implies 2 = a^2 + b^2n = c^2 + d^2n \geq 3$, a contradiction. So 2 is irreducible in $\mathbb{Z}[i\sqrt{n}]$.

Now recall that $n \geq 3$ is odd and consider $x = (1 + i\sqrt{n}), y = (1 - i\sqrt{n})$.

$$xy = (1 + i\sqrt{n})(1 - i\sqrt{n}) = 1 + n \text{ is an even integer} \implies 2 \mid xy$$

But $2 \nmid x = 1 + i\sqrt{n}$ and $2 \nmid y = 1 - i\sqrt{n}$. So we have found $x, y \in \mathbb{Z}[i\sqrt{n}]$ such that $2 \mid xy$ and $2 \nmid x$ and $2 \nmid y \implies 2$ is not prime in $\mathbb{Z}[i\sqrt{n}]$.

Therefore 2 is irreducible but not prime in $\mathbb{Z}[i\sqrt{n}]$.

(b) Set $n = 5$. From part (a), 2 is irreducible but not prime in $\mathbb{Z}[i\sqrt{5}]$. In a unique factorization domain, every irreducible element is prime. Therefore $\mathbb{Z}[i\sqrt{5}]$ must not be a unique factorization domain. $\square$

Problem 2. Let R be an integral domain and let $K = Q(R)$ be the field of fractions of R . For each maximal ideal $\mathfrak{m}$ of R , recall that $R_{\mathfrak{m}}$ denotes the localization of R at $\mathfrak{m}$, that is, $R_{\mathfrak{m}} = \left\{ \frac{x}{y} \in K \mid x \in R, y \in R \setminus \mathfrak{m} \right\}$. Let $\text{Max}(R)$ denote the set of maximal ideals of R . Prove that $R = \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}}$.

Proof. Recall that all maximal ideals are proper, so $1 \notin \mathfrak{m}, \forall \mathfrak{m} \in \text{Max}(R)$. Therefore, $\forall r \in R, \frac{r}{1} = r \in R_{\mathfrak{m}}, \forall \mathfrak{m} \in \text{Max}(R) \implies r \in \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}} \implies R \subseteq \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}}$.

On the other hand consider any $z \in \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}}$. and suppose $z \notin R$. Then let $I = \{r \in R \mid rz \in R\}$. Quickly, $\forall a, b \in I, \forall r \in R$,

$$\begin{aligned} az, bz \in R &\implies az - bz = (a - b)z \in R \implies a - b \in I \implies I \leq_+ R \\ r(az) = (ra)z \in R &\implies ra \in I \implies I \trianglelefteq R. \end{aligned}$$

Well, $1 \in I \implies 1z = z \in R$, a contradiction. So $1 \notin I \implies I$ is a proper ideal and therefore contained in some maximal ideal $\mathfrak{m}_z \in \text{Max}(R)$. Next, $z \in \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}} \implies z \in R_{\mathfrak{m}_z}$ so there exist $x \in R$ and $y \in R \setminus \mathfrak{m}_z$ such that $z = \frac{x}{y} \in R_{\mathfrak{m}_z}$ and $yz = x \in R$. So by definition, $y \in I \subseteq \mathfrak{m}_z$. But then $y \notin R \setminus \mathfrak{m}_z$, a contradiction. Therefore $z \in R$ and $\bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}} \subseteq R$.

Thus,

$$R = \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}}.$$

□

Problem 3. (a) Let R be a principal ideal domain and M a finitely generated projective R -module. Prove that M is a free R -module.

(b) Give an example showing that part (a) is false without assuming that R is a principal ideal domain.

Proof. (a) By an equivalence theorem, since M is projective, it is a direct summand of a free module F over R ; $M \oplus K \cong_R F$. Let ψ be this isomorphism and then let $\iota \in \text{Hom}_R(M, M \oplus K)$ be the natural inclusion defined by $m \mapsto (m, 0)$. This inclusion is injective, so it is an isomorphism to its image, which is $\text{Im}(\iota) = M \oplus 0$. Refer to the codomain restricted map, which is an isomorphism, as ι_M . Then $\psi \circ \iota_M$ is an isomorphism to $\psi(M \oplus 0)$ and $M \cong_R (\psi \circ \iota_M)(M) \leq_R F$, a submodule of the free module F .

Since any submodule of a free module over a PID is also free, M must be isomorphic to a free module, and therefore free itself over R .

(b) We give a counterexample to show not M need not be free if R isn't a PID.

Let $R = \mathbb{Z}_6$. This isn't even an integral domain since $[2]([3]) = [0]$ so it's not a PID. Now consider the $\mathbb{Z}_6$ -action $[k]_6(m) = [k]_n m$ on $\mathbb{Z}_n$ where $n \mid 6$. Quickly, for any $a, b, m \in \mathbb{Z}_n$ and any $[k], [c] \in \mathbb{Z}_6$,

$$\begin{aligned} \text{[Well-defined]} : [k]_6 = [k']_6 &\implies 6 \mid k - k' \implies n \mid k - k' \implies [k]_n = [k']_n \implies [k]_n m = [k']_n m \\ [k]_6(a + b) &= [k]_n(a + b) = [k]_n a + [k]_n b = [k]_6(a) + [k]_6(b) \\ [c]_6([k]_6(m)) &= [c]_6([k]_n m) = [c]_n[k]_n m = [ck]_n m = [ck]_6(m) \end{aligned}$$

So $\mathbb{Z}_n$ with $n \mid 6$ is an R -module. Now let $M = \mathbb{Z}_2$ and $K = \mathbb{Z}_3$, and note that both are R -modules since $2, 3 \mid 6$. finally consider $\langle (1, 1) \rangle \leq_+ \mathbb{Z}_2 \oplus \mathbb{Z}_3$. The order of $(1, 1)$ is $\text{lcm}(|1|_2, |1|_3) = 6$ and $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ has order 6 so in fact $M \oplus K = \mathbb{Z}_2 \oplus \mathbb{Z}_3$ is isomorphic to $R = \mathbb{Z}_6$ as a group and this is an R -isomorphism since both components are R -modules. Obviously R is a free module over itself via $R = \langle 1 \rangle_R$, so M is a direct summand of a free R -module and therefore projective by our equivalence theorem.

However, M is not free as an R -module since if it were $M \cong_R R^N \implies |M| = |R|^N \implies 2 = 6^N$ for some $N \in \mathbb{Z}^+$, which is impossible. So $M = \mathbb{Z}_2$ is a finitely generated projective R -module but not free over $R = \mathbb{Z}_6$ $\square$

Problem 4. Let R be a commutative ring, I an ideal of R , and M an R -module. Let IM be the submodule of M generated by the set $\{im \mid i \in I, m \in M\}$. Prove that we have the following isomorphism of R/I -modules: $(R/I) \otimes_R M \cong M/IM$.

Proof. Define $\phi : R/I \times M \rightarrow M/IM$ via $([r]_I, m) \mapsto [rm]_{IM}$. Note that

$$[r]_I = [s]_I \implies r - s \in I \implies rm - sm = (r - s)m \in IM \implies [rm]_{IM} = [sm]_{IM}$$

So ϕ is well-defined. We show this mapping is R -bilinear.

$$\begin{aligned} [+_L] : \phi([r_1]_I + [r_2]_I, m) &= \phi([r_1 + r_2]_I, m) = [(r_1 + r_2)m]_{IM} = [r_1m]_{IM} + [r_2m]_{IM} \\ &= \phi([r_1]_I, m) + \phi([r_2]_I, m) \\ [+_R] : \phi([r]_I, m_1 + m_2) &= [r(m_1 + m_2)]_{IM} = [rm_1 + rm_2]_{IM} = [rm_1]_{IM} + [rm_2]_{IM} \\ [\smile] : \phi([r]_I, sm) &= [(rs)m]_{IM} = [r(sm)]_{IM} = \phi([r]_I, sm) = [r(sm)]_{IM} \\ &= s[rm]_{IM} = s\phi([r]_I, m) \end{aligned}$$

By **Universal Property of Tensors** we have a unique R -module homomorphism $\Phi : R/I \otimes_R M \rightarrow M/IM$ defined by $\Phi([r]_I \otimes m) = [rm]_{IM}$. We show this mapping is bijective by constructing an inverse.

Let $\Phi^{-1} : M/IM \rightarrow (R/I) \otimes_R M$ be defined as follows. For each $[m]_{IM} \in M/IM$, choose a representative $m \in M$ and set $\Phi^{-1}([m]_{IM}) = [1]_I \otimes m$. We show this mapping is well-defined. If $[m]_{IM} = [m']_{IM}$, then $m - m' \in IM$. So then $m - m' = \sum_{k=1}^n i_k m_k$ for some $i_1, \dots, i_n \in I$ and some $m_1, \dots, m_n \in M$. Therefore,

$$([1]_I \otimes m) - ([1]_I \otimes m') = [1]_I \otimes \left(\sum_{k=1}^n i_k m_k \right) = \sum_{k=1}^n [1]_I \otimes i_k m_k = \sum_{k=1}^n [i_k]_I \otimes m_k = \sum_{k=1}^n [0]_I \otimes m_k = 0.$$

So Φ^{-1} is well-defined. Finally, $\Phi(\Phi^{-1}([m]_{IM})) = \Phi([1]_I \otimes m) = [m]_{IM}$ and then on the other hand $\Phi^{-1}(\Phi([r]_I \otimes m)) = \Phi^{-1}([rm]_{IM}) = [1]_I \otimes rm = [r]_I \otimes m$. So then Φ is an isomorphism and $(R/I) \otimes_R M \cong M/IM$.

□

Problem 5. Let R be an integral domain. Prove that if R is injective as an R -module, then R is a field.

Proof. By **Baer's Criterion** if R is injective as an R -module, then for every ideal $I \trianglelefteq R$, and any $h \in \text{Hom}_R(I, R)$, there exists $f \in \text{Hom}_R(R, R)$ such that $f \circ \iota_I = h$ where ι_I is the natural inclusion from I to R , an injective R -module homomorphism.

For any non-zero $x \in R$, set $I = xR$. Quickly, $a, b \in I \implies a = x\alpha, b = x\beta$ for some $\alpha, \beta \in R \implies a - b = x\alpha - x\beta = x(\alpha - \beta) \in xR$ and $r(a) = rx\alpha = x(r\alpha) \in xR \implies I = xR$ is an ideal of R . Now let $h : I \rightarrow R$ be defined by $h(xr) = r$. $xr = xr' \in I \implies xr - xr' = x(r - r') = 0 \implies r - r' = 0 \implies r = r'$ since R is an integral domain, so h is well-defined. Next we show that $h \in \text{Hom}_R(I, R)$ is surjective.

$$\begin{aligned} [+]: h(xr + xs) &= h(x(r + s)) = r + s = h(xr) + h(xs) \\ [\curvearrowright]: sh(xr) &= sr = h(x(sr)) = h(s(xr)) \\ [\twoheadrightarrow]: \forall r \in R, & h(xr) = r \end{aligned}$$

Now, since R is an injective R -module, there exists some $f \in \text{Hom}_R(R, R)$ such that $f \circ \iota_I = h$. Finally, since $h(x \cdot 1) = h(x) = 1$, $(f \circ \iota_I)(x) = f(x) = x(f(1)) = 1$ and so we see that $f(1) = x^{-1}$ since R is commutative. So any non-zero $x \in R$ has an inverse and since R is already an integral domain, it must be a field.

□

Problem 6. Let $A = \begin{pmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & 0 & -2 \end{pmatrix} \in M_3(\mathbb{R})$.

(a) Find the Jordan canonical form of A .

(b) For what values of $a, b \in \mathbb{R}$ is A diagonalizable?

Proof. (a) $|A - I\lambda| = (3 - \lambda) \begin{vmatrix} 3 - \lambda & 0 \\ 0 & -(2 + \lambda) \end{vmatrix} = (3 - \lambda)(-(3 - \lambda)(2 + \lambda)) = -(3 - \lambda)^2(2 + \lambda) = 0$
 $\implies \lambda_1 = 3, \lambda_2 = -2$ are the eigenvalues of A with algebraic multiplicities 2 and 1, respectively. Since $\lambda_2 = -2$ has algebraic multiplicity 1, that must also be its geometric multiplicity and so the Jordan block it contributes has size 1 and then must be (-2) . Next we compute the geometric multiplicity $\text{nullity}(A - 3I)$ of $\lambda_1 = 3$.

$$A - 3I = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & 0 & -5 \end{pmatrix} \text{ so } \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & 0 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies \begin{cases} ax = 0 \\ bx - 5z = 0 \end{cases}.$$

$ax = 0 \implies a = 0$ or $x = 0$ since $\mathbb{R}$ is an integral domain. So we have two cases here: (i) and (ii).

(i) If $a = 0$, then x, y are free variables and $z = \frac{b}{5}x$ which implies that all vectors of the form $\begin{pmatrix} x \\ y \\ \frac{b}{5}x \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ \frac{b}{5} \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ are eigenvectors of $\lambda_1 = 3 \implies \text{nullity}(A - \lambda_1 I) = 2$ and so λ_1 contributes two Jordan blocks of total size 2, and so they must both be of size 1. That is, $J_1 = J_2 = (3)$, $J_3 = (-2)$ and the Jordan canonical form of A is

$$J(A) = J_1 \oplus J_2 \oplus J_3 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

(ii) If $a \neq 0$, then $x = 0$ and so $z = \frac{b}{5}x \implies z = 0$ and y is our only free variable. That is, $\text{nullity}(A - \lambda_1 I) = 1$. So then λ_1 must contribute one Jordan block of size 2. That is, $J_1 = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ and λ_2 contributes the Jordan block $J_2 = (-2)$ and so the Jordan canonical form of A is

$$J(A) = J_1 \oplus J_2 = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

(b) A matrix is diagonalizable if and only if every Jordan block in it has size 1. By (a) this only happens when $a = 0$. There is no restriction on b .

□

Problem 7. Let $(a, b) \in \mathbb{Z}^2$. Show that there exists a basis of the free $\mathbb{Z}$ -module $\mathbb{Z}^2$ containing (a, b) if and only if a and b are relatively prime.

Proof. ($\implies$) Suppose that there exists a basis of $\mathbb{Z}^2$ containing (a, b) . Then there is some $(c, d) \in \mathbb{Z}^2$ linearly independent from (a, b) such that $\mathbb{Z}^2 = \langle (a, b), (c, d) \rangle_{\mathbb{Z}}$. So then the matrix $((a, b)^T \mid (c, d)^T)$ has rank 2. That is,

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \text{ is invertible over } \mathbb{Z} \implies \frac{1}{ad - bc} \in \mathbb{Z} \implies ad - bc \text{ is a unit in } \mathbb{Z}.$$

Well, then if δ is a common divisor of a and b , it must also divide $ad - bc = \pm 1 \implies \delta = \pm 1 \implies \gcd(a, b) = 1$. So a and b are relatively prime.

($\impliedby$) On the other hand if a and b are relatively prime, then by Bézout's Identity there exist $x, y \in \mathbb{Z}$ such that $ax + by = 1$. So let $(c, d) = (-y, x)$. Then

$$\det \begin{pmatrix} a & -y \\ b & x \end{pmatrix} = ax + by = 1 \implies \begin{pmatrix} a & -y \\ b & x \end{pmatrix}^{-1} = \frac{1}{ax + by} \begin{pmatrix} x & y \\ -b & a \end{pmatrix} = \begin{pmatrix} x & y \\ -b & a \end{pmatrix}$$

Since matrix is invertible over $\mathbb{Z}$ it has rank 2, it's columns are linearly independent over $\mathbb{Z}$ and therefore form a basis for $\mathbb{Z}^2$ over $\mathbb{Z}$. That is, $\langle (a, b), (-y, x) \rangle_{\mathbb{Z}} = \mathbb{Z}^2$ over $\mathbb{Z}$.

□